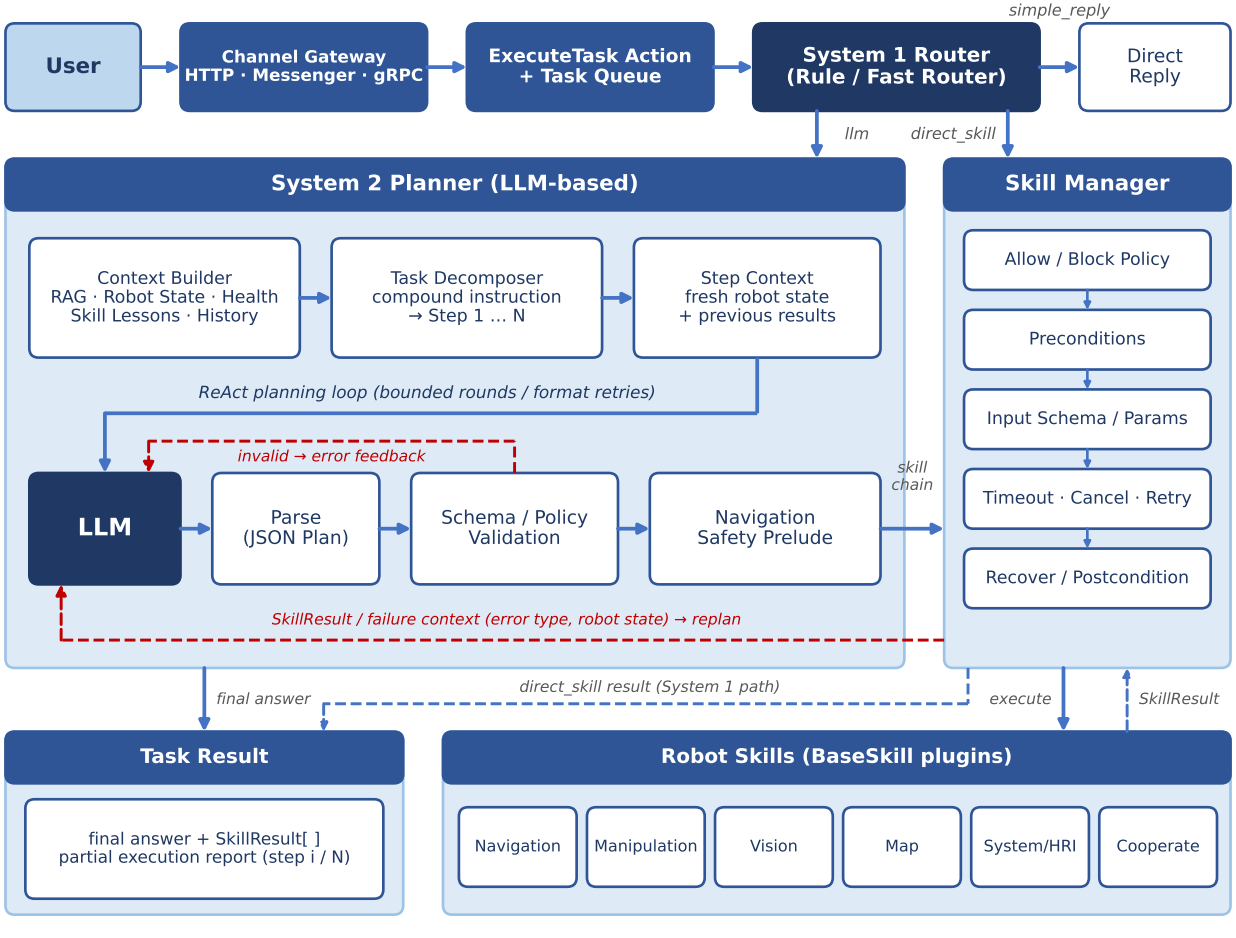
User
Channel Gateway
HTTP · Messenger · gRPC
ExecuteTask Action
+ Task Queue
System 1 Router
(Rule / Fast Router)
simple_reply
Direct
Reply
llm
direct_skill
System 2 Planner (LLM-based)
Skill Manager
Context Builder
RAG · Robot State · Health
Skill Lessons · History
Task Decomposer
compound instruction
→ Step 1 … N
Step Context
fresh robot state
+ previous results
ReAct planning loop (bounded rounds / format retries)
invalid → error feedback
LLM
Parse
(JSON Plan)
Schema / Policy
Validation
Navigation
Safety Prelude
skill
chain
Allow / Block Policy
Preconditions
Input Schema / Params
Timeout · Cancel · Retry
Recover / Postcondition
SkillResult / failure context (error type, robot state) → replan
final answer
direct_skill result (System 1 path)
execute
SkillResult
Task Result
final answer + SkillResult[ ]
partial execution report (step i / N)
Robot Skills (BaseSkill plugins)
Navigation
Manipulation
Vision
Map
System/HRI
Cooperate